

*MA.7.AR.4.3***Benchmark**

MA.7.AR.4.3 **Given a mathematical or real-world context, graph proportional relationships from a table, equation or a written description.**

Benchmark Clarifications:

Clarification 1: Instruction includes equations of proportional relationships in the form of $y = px$, where p is the constant of proportionality.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.3.3
- MA.8.AR.2.1
- MA.8.AR.3.4
- MA.8.AR.4.2, MA.8.AR.4.3
- MA.8.DP.1.3

Terms from the K-12 Glossary

- Constant of Proportionality
- Proportional Relationships
- Unit Rate

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.4, MA.6.AR.3.5
- MA.6.GR.1.1
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3

Next Benchmarks

- MA.912.AR.2.4

Purpose and Instructional Strategies

In previous courses, students solved problems involving ratios, rates and unit rates, and began plotting points on the coordinate plane. In grade 7 accelerated, students begin working with proportional relationships, including graphing proportional relationships given a table, equation or written description. Students will also determine if a linear relationship is proportional and graph two-variable linear equations from a written description, table or equation in slope-intercept form. In Algebra 1, students graph linear equations in other forms, as well as from tables and written descriptions.

- Instruction includes different ways of representing proportional relationships, such as tables, equations, and graphs. Multiplying or dividing one quantity in a ratio by a particular factor requires doing the same with the other quantity in the ratio to maintain the proportional relationship. Graphing equivalent ratios create a straight line passing through the origin. The equations generated with the ratios will be unique in that they will follow the form of $y = px$.

- Tables

<i>x</i>	5	40	65	
<i>y</i>	1	8	13	

- Equations

$$y = \frac{1}{5}x$$

- Given one representation, have students provide the other three in order to determine their mastery of these equivalencies (as noted in MA.7.AR.4.4).
 - For example, if given a written description, have students provide a table, an equation and a graph of the given proportional relationship.
- Students should understand that although the most common variable used to represent the constant of proportionality is p , any other variable can be used.
 - For example, students can write the equation $p = 1.6$ or $k = 1.6$ to state that the constant of proportionality is 1.6 given the equation $y = 1.6x$.
- When an equation or written description is given, have students create a corresponding table of values to assist with graphing. Be sure to emphasize the use of the origin as one of the points for the table (*MTR.2.1*).

Common Misconceptions or Errors

- Students may confuse the dependent and independent variables when graphing. To address this conception, instruction includes the understanding that the independent variable depends on the given context. Additionally, independent variables are not always the x -axis and the dependent variables are not always the y -axis.
- For example, if a student has a proportional relationship between feet and meters, they can graph feet either on the x -axis or the y -axis. The dependent one depends on the context. For instance, if one is given feet and converting to meters, then feet would be independent, and meters would be dependent.
- Students may confuse the x - and y -axis.
- Students may not recognize the appropriate axis scale to graph the given scenario efficiently.
 - For example, if a situation involves fractional numbers, using a scale of 1 may not be appropriate. Instead, students should consider using a scale with a fractional value.

Strategies to Support Tiered Instruction

- Instruction focuses on students' comprehension of the context or situation by engaging in questions (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - What do you know from the problem?
 - What is the problem asking you to find?
 - What are the two quantities in this problem?
 - How are the quantities related to each other?
 - Which quantity do you want to consider as the independent variable?
 - Which quantity do you want to consider as the dependent variable?

- Instruction includes co-creating a graphic organizer with key features of coordinate plane such as x -axis, y -axis, origin, quadrants, scales, origin and coordinates.
- Instruction includes the understanding that the independent variable depends on the given context. Additionally, independent variables are not always the x -axis and the dependent variables are not always the y -axis.
 - For example, if one has a proportional relationship between feet and meters, students can graph feet either on the x -axis or the y -axis. The dependent variable depends on the context. For instance, if one is given feet and converting to meters, then feet would be independent, and meters would be dependent.
- Teacher provides instruction on creating a table of values to assist graphing an equation on a coordinate plane.

x	$y = 2x$	y	Coordinate Points
-2	$y = 2(-2)$ $y = -4$	-4	$(-2, -4)$
0	$y = 2(0)$ $y = 0$	0	$(0, 0)$
6	$y = 2(6)$ $y = 12$	12	$(6, 12)$

- Instruction includes modeling how to properly use a scale when situations involve fractional numbers. Instead of using a scale of 1, use a scale with a fractional value.

Instructional Tasks

Instructional Task 1 (MTR.5.1)

Coffee costs \$18.96 for 3 pounds at the store, CoffeeUs.

Part A. What is the cost per pound of coffee?

Part B. At CoffeeUs, the price for a pound of coffee is the same no matter how many pounds you buy. Let x be the number of pounds of coffee and y be the total cost of x pounds. Draw a graph of the proportional relationship between the number of pounds of coffee and the total cost.

Part C. Where can you see the cost per pound of coffee in the graph? What is it?

Instructional Items

Instructional Item 1

The cost of Hass avocados is a proportional relationship to the number of avocados being purchased. The equation $c = 2.10a$ represents this relationship where c is the total cost and a is the number of Hass avocados being purchased. Create a graph representing this relationship.

Instructional Item 2

Graph the proportional relationship given in the table below.

<i>x</i>	1	2	4	5
<i>y</i>	$\frac{5}{2}$	5	10	$\frac{25}{2}$

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*